Nonstationary Time Series Models

Rebecca Killick(r.killick@lancs.ac.uk)
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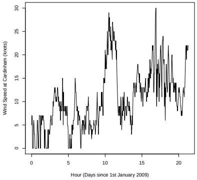
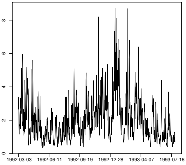
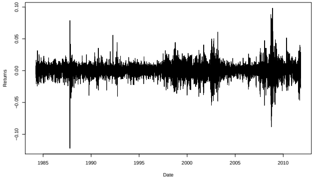
Nonstationary TS

Statistical properties of a series are **NOT** constant over time.

These arise in a variety of applications including;







Wind Speed | Wave Heights | Financial Returns.

What do I mean?

- I mean *NOT* second-order stationary.
- So, unconditional variance *changes* with time.
- Autocovariance, spectrum, etc. *change* with time.
- Typically assume $\mathcal{E}X_t = 0$ (assume mean removed).

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More interested in Variability and CI

Structure

- Motivation
- Overview of Nonstationary Time Series Models
- The Trend Locally Stationary Wavelet Process
- Examples
- Forecasting Stationary TS reminder
- Nonstationary forecasting
- Practical examples

Motivation

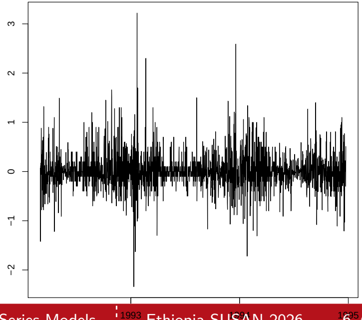
Motivation

Time series are important in energy applications e.g.

- short-term wind speed forecasting,
- predicting future energy use for groups of customers,
- planning of proposed wind farms.

Our industrial collaborators are interested in forecasting such processes.

- Data are often assumed to be (second order) stationary.
- Many series arising from energy applications are inherently nonstationary!



Motivation

- It is common for practitioners to forecast their nonstationary time series using tools from the stationary world.
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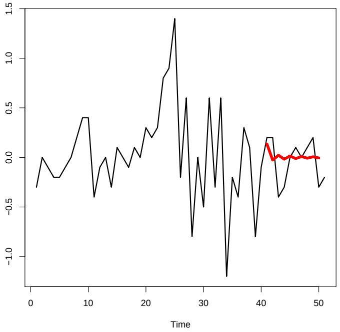
- Red - Full Series estimation.



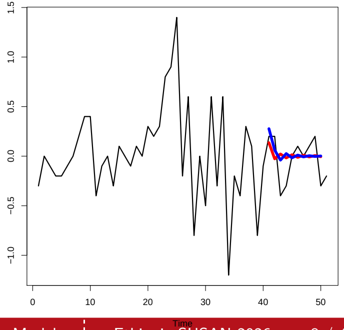
Motivation

- It is common for practitioners to forecast their nonstationary time series using tools from the stationary world.

- What might be the danger here?

- Red - Full Series estimation.
- Blue - Last 30 observations.

- Both fit ARMA(2,2) and quickly decay to 0 estimates.



Motivation

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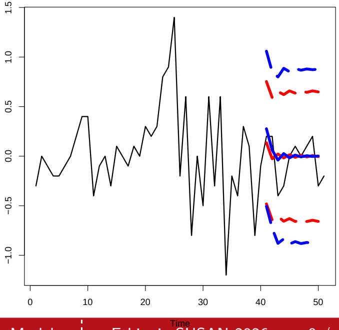
- What might be the danger here?

- Red - Full Series estimation.
- Blue - Last 30 observations.

- Both fit ARMA(2,2) and quickly decay to 0 estimates.
- Overconfidence in forecast?



Overview of Nonstationary Time Series Models

Models

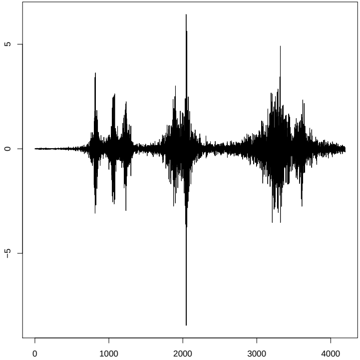
Modelling in the face of non-stationarity is no easy task!

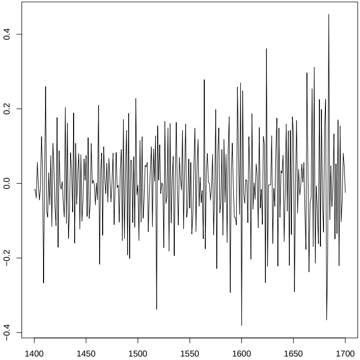
- Various approaches have been explored, built on models that fit particular types of non-stationarity:
  - assume piecewise stationarity (changepoints from previous talk);
  - use parametric models with time-changing coefficients, e.g. Time Varying AR (tvAR).

- Processes with a slowly time-varying second order structure are known as locally stationary (LS).
  - Advanced LS models (ARCH) (Dahlhaus and Subba Rao, 2006).
  - Locally stationary fourier processes (Dahlhaus, 1997).
  - Locally stationary wavelet processes (Nason et al., 2000).

Locally stationary

If you take a small enough region, it will appear stationary as the structure varies slowly over time.





Application areas include;

- medicine, finance
- environmental processes, e.g. wind speeds.

LS Fourier

- Traditional stationary Fourier:

$$X_t = \int_{-\pi}^{\pi} A(\omega) \exp(i\omega t)\, \mathrm{d}\xi(\omega)$$

LS Fourier

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Locally stationary Fourier (LSF) processes:

$$X_{t,T} = \int_{-\pi}^{\pi} A_{t;T}(\omega) \exp(i\omega t)\, \mathrm{d}\xi(\omega)$$

- Processes with a slowly time varying second-order structure can be modelled using the traditional sine waves, but also using wavelets.

Trend Locally Stationary Wavelet Processes

T-LSW model

A nonstationary time series can be modelled as a trend locally stationary wavelet (T-LSW) process $\{X_{t,T}\}$, $t = 0, \ldots, T-1$, as follows:

$$X_{t,T} = \mu(t/T) + \sum_{j=-\infty}^{-1} \sum_{k \in \mathbb{Z}} w_j(k/T)\psi_{j,k-t}\xi_{j,k},$$

where

- μ is a low order polynomial mean function,
- $w_j(k/T)$ are amplitudes, the smoothness of which controls the degree of nonstationarity,
- $\{\psi_{j,k-t}\}_{j,k}$ is a set of discrete non-decimated wavelets,
- $\{\xi_{j,k}\}$ is a sequence of orthonormal random variables.

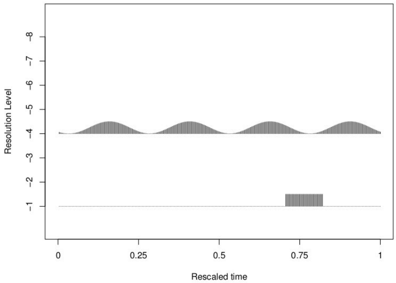
McGonigle et al. (2022) JTSA, `TLSW` package on CRAN

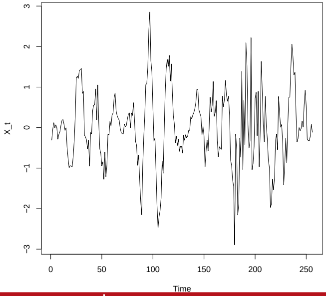
The wavelet spectrum is defined as:

$$S_j(z) = \lim_{T\to\infty} |w_{j,\lfloor zT \rfloor;T}|^2, \text{ where } z \in (0,1).$$

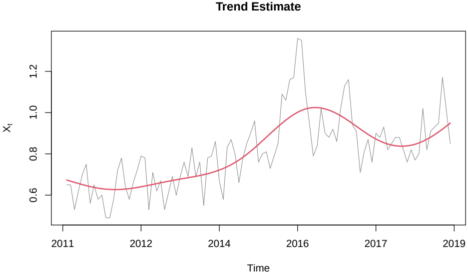
- Contribution to process variance at scale j and re-scaled time z.
- Estimates are biased and inconsistent – we must correct and smooth.
- Correspondence between the spectrum and the acf,

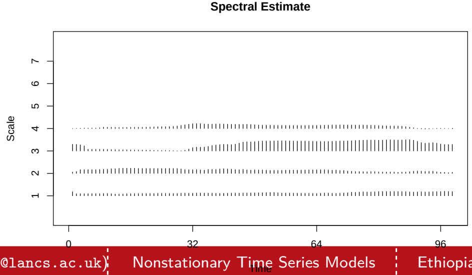
$$c(z,\tau) = \sum S_j(z)\Psi_j(\tau).$$





Example TLSW





Local autocovariance

The local autocovariance (LACV) is given by

$$c(z,\tau) = \sum_j S_j(z)\Psi_j(\tau),$$

- $\Psi_j(\tau)$ is the autocorrelation wavelet at scale j, lag τ
- $S_j(z)$ is the spectrum at rescaled time z

We estimate this using

$$\hat{c}(z,\tau) = \sum_{j=-J_0}^{-1} \hat{S}_j(z)\Psi_j(\tau),$$

where $J_0 = \alpha\lfloor \log_2(T) \rfloor$ for $\alpha \in (0,1)$ is the coarsest scale used.

Local pacf

We define the lpacf at time t and lag τ as:

$$q(t,\tau) = \text{Corr}(X_t, X_{t-\tau} | \{X_{t-1} \ldots, X_{t-\tau+1}\})$$

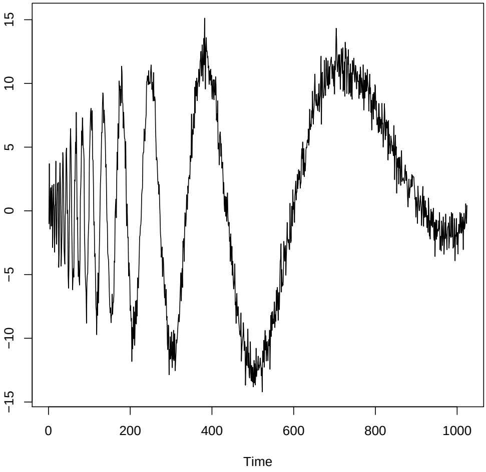
$$q_T(t,\tau) = \begin{cases} \frac{c\left(\frac{t-1}{T},1\right)}{\sqrt{c\left(\frac{t}{T},0\right)c\left(\frac{t-1}{T},0\right)}}, & \text{for } \tau = 1 \\ \varphi_{t,\tau,\tau}\sqrt{\frac{\text{MSPE}(\hat{x}_t, X_t | X_{t-1} \ldots, X_{t-\tau+1})}{\text{MSPE}(\hat{x}_{t-\tau}, X_{t-\tau} | X_{t-1} \ldots, X_{t-\tau+1})}}, & \text{for } \tau \geq 2. \end{cases}$$

For stationary models the square root equals 1 and the $\varphi_{t,\tau,\tau}$ is the usual estimate of the pacf.

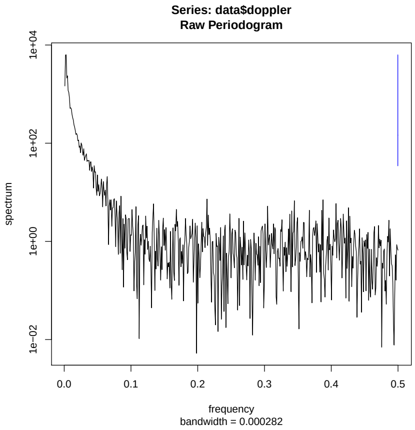
Theoretical treatment given in Killick et al. (2020) EJS, `lpacf` on CRAN

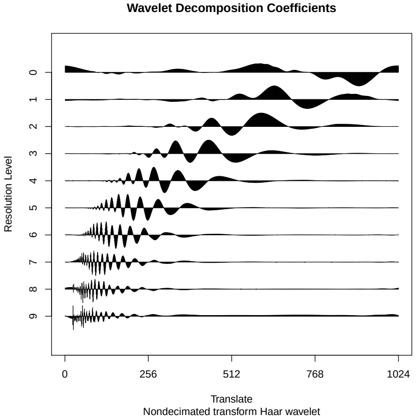
Examples

Doppler

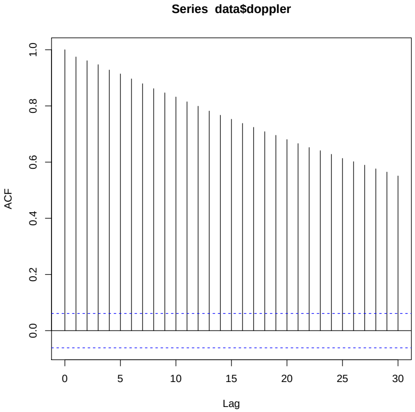


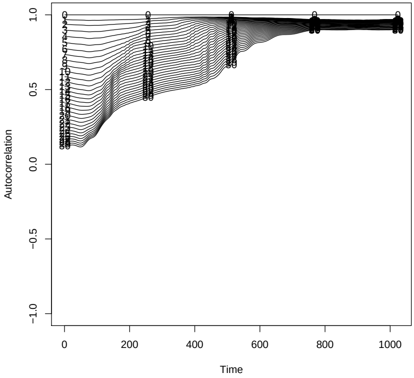
Doppler


Series: data$doppler
Raw Periodogram
spectrum
1e+04
1e+02
1e+00
1e−02
0.0
0.1
0.2
0.3
0.4
0.5
frequency
bandwidth = 0.000282


Wavelet Decomposition Coefficients
Resolution Level
0
1
2
3
4
5
6
7
8
9
0
256
512
768
1024
Translate
Nondecimated transform Haar wavelet

Doppler


Series data$doppler
ACF
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30
Lag


Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
0
200
400
600
800
1000
Time

Time-Varying AR(1)

Model is time-varying autoregressive of order 1.

$$X_t = \alpha_t X_{t-1} + Z_t,$$

where $Z_t \sim N(0, \sigma^2)$ iid.

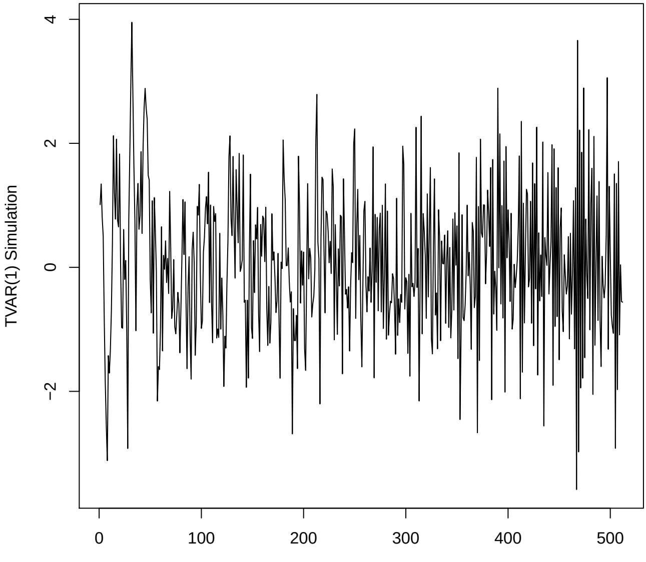
Here $t = 1, \ldots, 512$.

Let α_t vary linearly from +0.9 to -0.9 over whole series.

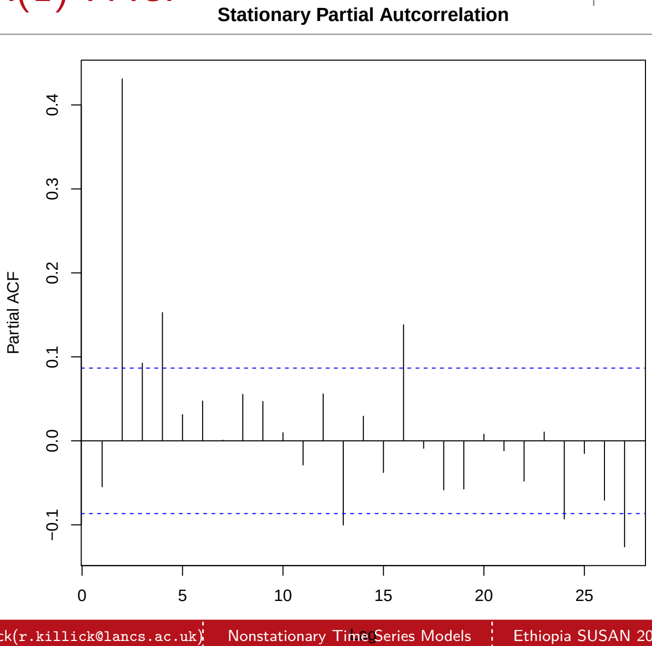
Would expect

- pacf at lag 1 to vary from +0.9 to -0.9
- pacf at lag 2 to be zero

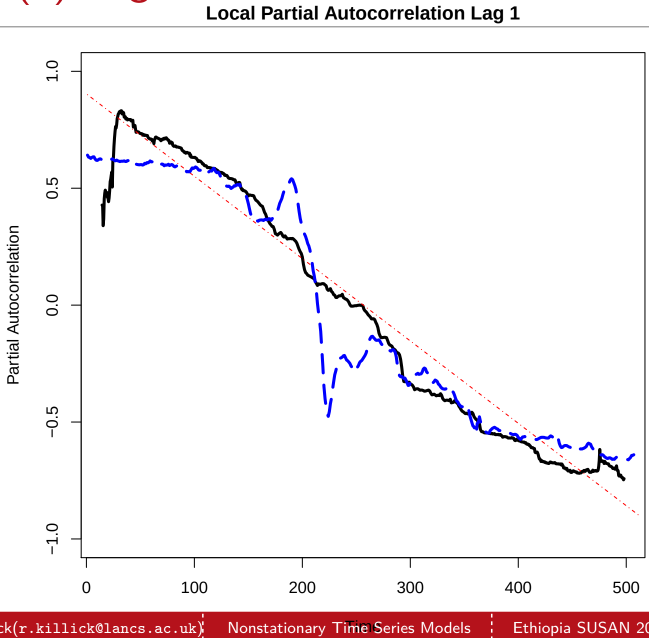
TVAR(1) simulation



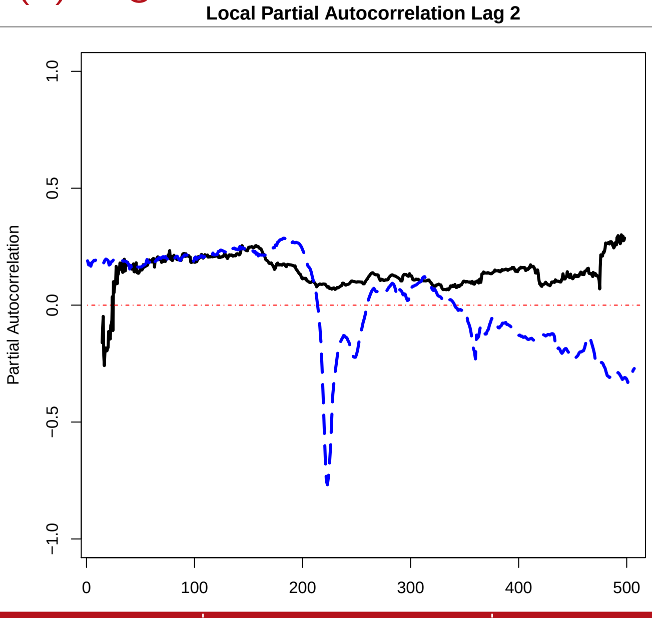
TVAR(1) PACF


Stationary Partial Autcorrelation
Partial ACF
0.4
0.3
0.2
0.1
0.0
−0.1
0
5
10
15
20
25
Lag

TVAR(1) Lag 1 LPACF



TVAR(1) Lag 2 LPACF



School of Mathematical Sciences | Lancaster University

Forecasting Nonstationary Time Series

Stationary Forecasting

Suppose have data $x_1, \ldots, x_T$ from *stationary* series.

Want to make forecast $\hat{x}_T(h)$ made at time T for horizon h.

Want forecast using linear combination of past:

$$\begin{aligned}\hat{x}_T(1) &= \sum_{i=0}^{T-1} \varphi_i x_{T-i} \\ &= \varphi_0 x_T + \varphi_1 x_{T-1} + \varphi_2 x_{T-2} + \cdots\end{aligned}$$

Note: φ sequence DOES NOT depend on time (stationary x_t)

Theory can tell us optimal least-square forecast (Box-Jenkins).

Stationary Forecasting

Stationary AR(p) model:

$$X_t = \sum_{i=1}^{p} \varphi_i X_{t-i} + \epsilon_t.$$

For forecasting we teach UG students to:

1. Decide on the most appropriate model for the data;
   - use the pacf to aid choice of p,
   - estimate the coefficients using MLE or Yule-Walker equations:

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{bmatrix} = \begin{bmatrix} \gamma_0 & \gamma_{-1} & \cdots \\ \gamma_1 & \gamma_0 & \cdots \\ \vdots & \vdots & \ddots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots \end{bmatrix} \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_p \end{bmatrix}$$

   where $\gamma_.$ is the autocovariance function for X_t.
2. Forecast the data using the fitted coefficients and setting $\epsilon_t = 0$.

Extension to Nonstationary Time Series

LSW Forecasting

Given observations $x_0, \ldots, x_{t-1}$, we:

- Predict $\hat{x}_t = \sum_{s=t-p}^{t-1} b_{t-1-i,T} X_s$, where

- p is the number of latest observations used for prediction and
- $\boldsymbol{b}$ is the solution to localized Yule-Walker equations.

$$\begin{bmatrix} c(t,1) \\ c(t,2) \\ \vdots \\ c(t,p) \end{bmatrix} = \begin{bmatrix} c(t-1,0) & c(t-2,-1) & \cdots \\ c(t-1,1) & c(t-2,0) & \cdots \\ \vdots & \vdots & \ddots \\ c(t-1,p) & c(t-2,p-1) & \cdots \end{bmatrix} \begin{bmatrix} b_{t-1} \\ b_{t-2} \\ \vdots \\ b_{t-1-p} \end{bmatrix}$$

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These require knowledge of the covariance structure at time t which is the point we are trying to predict.

- The covariance at time t is similar to that at time $t-1$.
- We extrapolate at the smoothing step.

LSW Forecasting

- We really want p to reflect *local* AR order
- This tells you how far back in time is relevant
- It mirrors the stationary approach

Thus we,

- use the lpacf to choose p and
- produce a data driven approach for practitioners.

Full details in Killick et al. (2025) JRSSC.

Forecasting Simulations

Elephant in the room

Stationary forecasting is hard to beat

Elephant in the room

Stationary forecasting is hard to beat

- Variance compensates for nonstationarity
- Can compensate with higher order models

Simulation Study

Range of models considered:

- TVAR(1)
- TVAR(2)
- TVAR(12)
- TVMA(1)
- TVMA(2)
- Uniformly modulated white noise
- LSW process

Summary

lpacf greatly improves on ARMA in 2/3 of cases
with comparable results (ratio ± 0.05) in remaining 1/3.

Forecast Wind Energy

- Vital for national energy security and economy
- Vital for suppliers
- Naturally non-stationary: succession of pressure systems?
- Many inputs used for forecasting — complex business
- E.g. simple direction variable can help . . .
- However, short term ARIMA forecasting often part of this.
- But this area is a small cog in big wheel!

Forecast Results

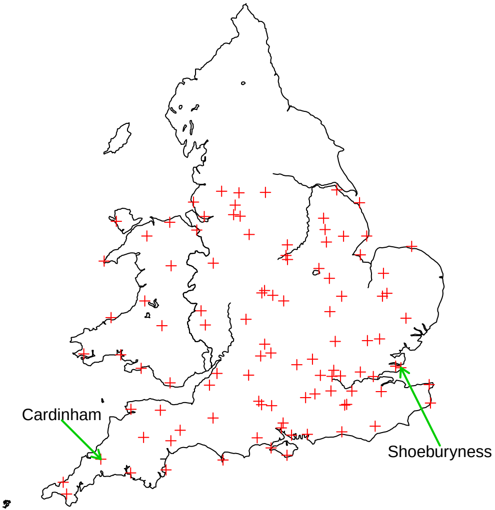
- Recall $\mathbb{E}X_t = 0$ assumption
- Hence, more interested in variability forecast
- Q: Does actual value lie in our prediction band?

- Ex: predict last observation from previous ones.
- Then predict second to last from previous ones.
- Then 3rd to last, 4th to last, etc.

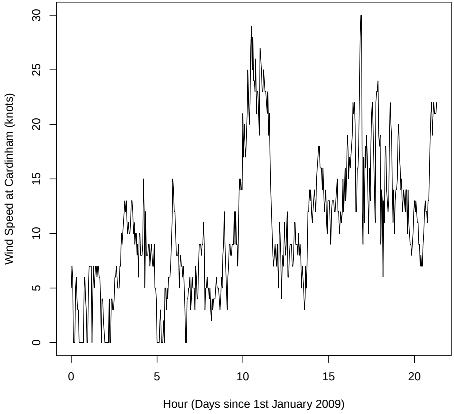
Data

Data is MIDAS obtained from British Atmospheric Data Centre originally from the UK Met Office.
Acknowledgement and thanks is due.

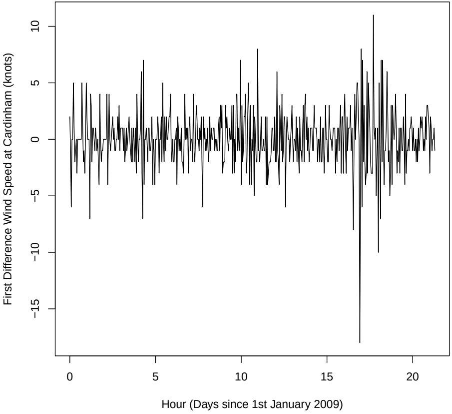
Locations



Cardinham



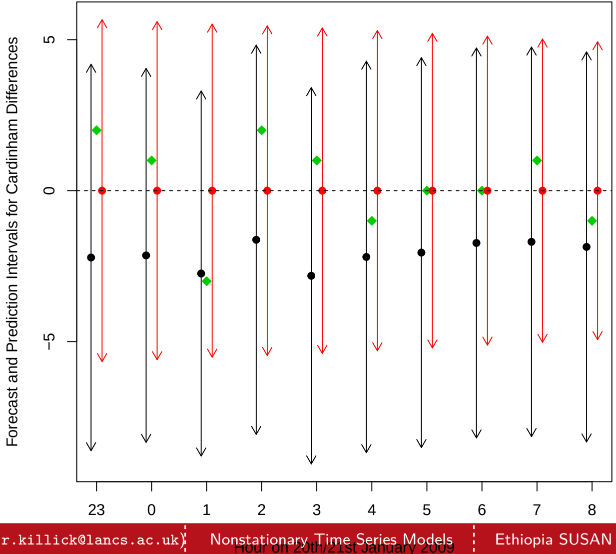
Wind Speed at Cardinham (knots)
0
5
10
15
20
25
30
0
5
10
15
20
Hour (Days since 1st January 2009)



First Difference Wind Speed at Cardinham (knots)
−15
−10
−5
0
5
10
0
5
10
15
20
Hour (Days since 1st January 2009)

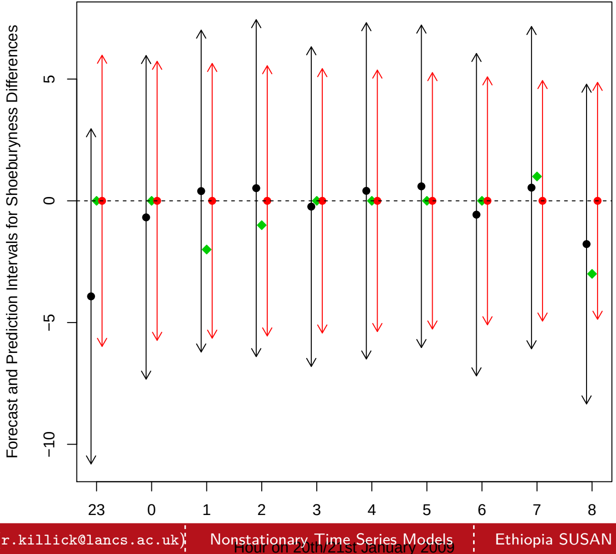
Cardinham

$MPE_s = 7.18, MPE_{ls} = 2.19,$ Red - lpacf Black - ARMA



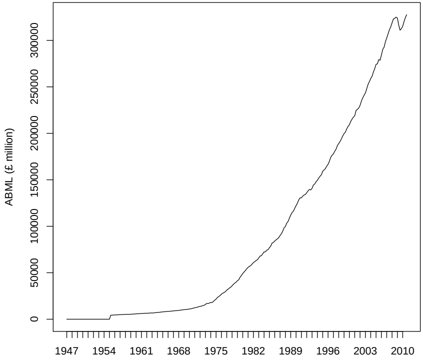
Shoeburyness

$MPE_s = 2.67, MPE_{ls} = 1.49,$ Red - lpacf Black - ARMA

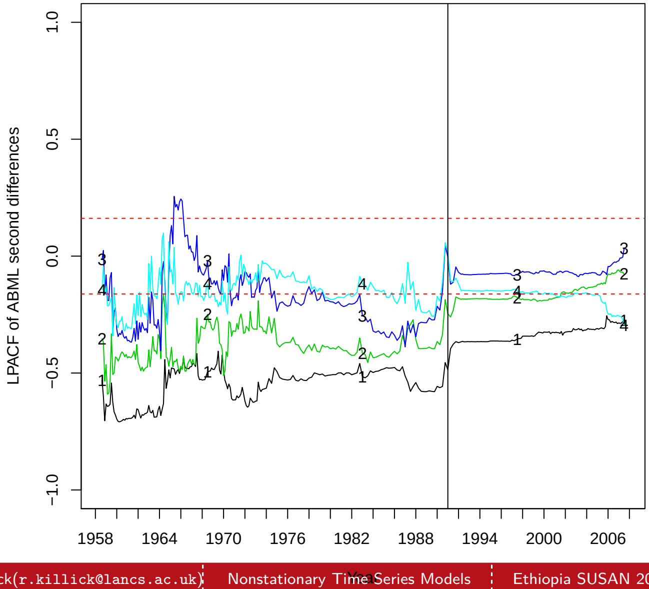


Motivation - ABML

- ABML consists of gross value added amounts
- Component in the estimate of GDP
- 223 observations from Q1 1955 to Q3 2010
- We use second differences to remove trend
- Tests of stationarity reject H0.



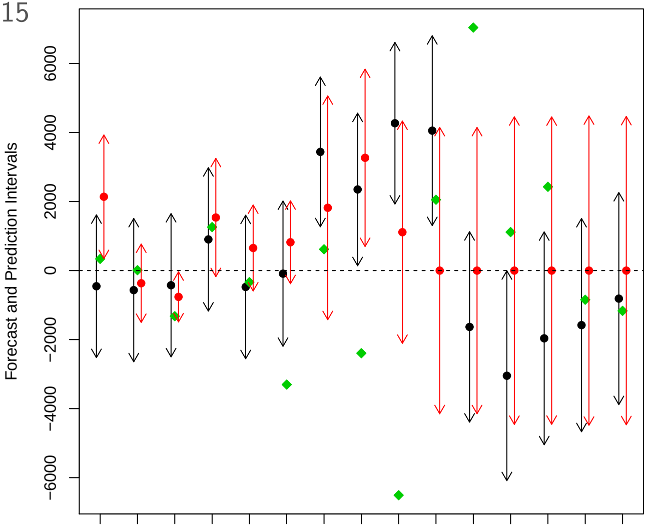
ABML lpacf



LPACF of ABML second differences
1.0
0.5
0.0
−0.5
−1.0
1958
1964
1970
1976
1982
1988
1994
2000
2006
Year

AMBML - lpacf forecast

RMSE: lpacf=£3430m, B-J=£4290m, Fryzlewicz=£8830m, 11/15, 8/15, 6/15



Summary

- Introduced the Trend Locally Stationary Wavelet process for modelling nonstationary time series and important quantities
- Motivated why forecasting nonstationary time series is important.
- Used the lpacf to choose p for the localised Yule-Walker equations.
- Showed increased forecasting performance when using the lpacf.

- We have used the lpacf as a tool for forecasting but it can be used in a variety of settings.

Software

The software for T-LSW processes is available in the `TrendLSW` R package on CRAN:
`http://cran.r-project.org/web/packages/TrendLSW`

A test for stationarity (and other useful functions) are available in the `locits` R package on CRAN:
`http://cran.r-project.org/web/packages/locits/`

The software for the LPACF and forecasting LSW processes is available in the `lpacf` and `forecastLSW` R packages.
`https://cran.r-project.org/web/packages/lpacf/`
`https://cran.r-project.org/web/packages/forecastLSW/`